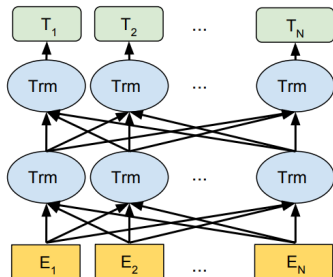


Deep Learning for NLP

BERT

ARLMs vs. MLM



Learning goals

- Understand the concept of self-supervision
- Gain ability to distinguish different types of language models

AGAIN: WHAT IS A LANGUAGE MODEL?

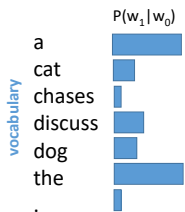
- Statistical model that predicts text that fits well for a given context (typically also text)
- Auto-regressive LMs (ARLMs)
 - Predict one word that is highly likely given a prompt (previous words)
 - For predicting an entire text, repeat the process (i.e., extend the prompt with previously predicted words)
 - To predict a text from scratch, use an extra symbol <START> as the initial prompt

ARLM: TOY EXAMPLE

<START>

w_0

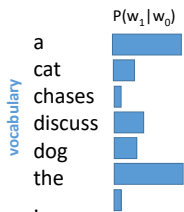
ARLM: TOY EXAMPLE



<START>

w_0

ARLM: TOY EXAMPLE



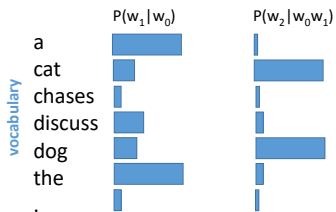
<START>

the

w_0

w_1

ARLM: TOY EXAMPLE



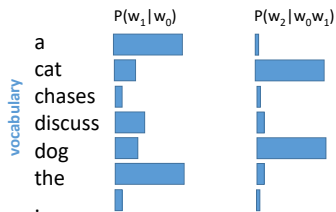
<START>

the

w_0

w_1

ARLM: TOY EXAMPLE



<START>

w_0

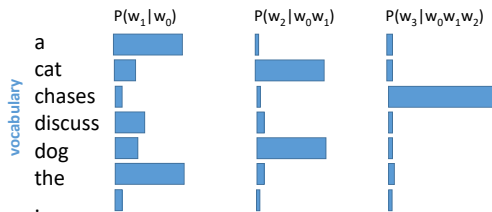
the

w_1

dog

w_2

ARLM: TOY EXAMPLE



<START>

w_0

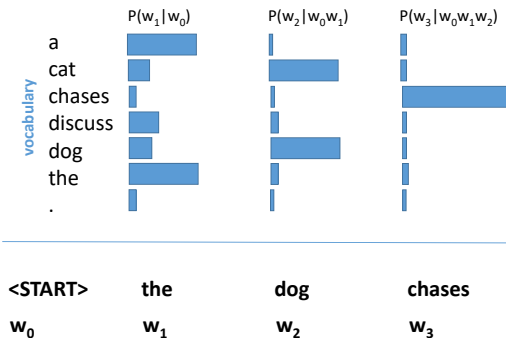
the

w_1

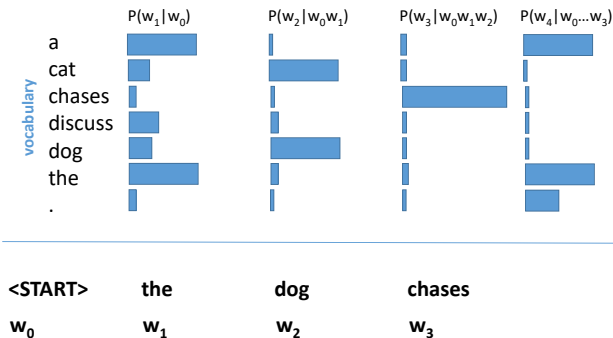
dog

w_2

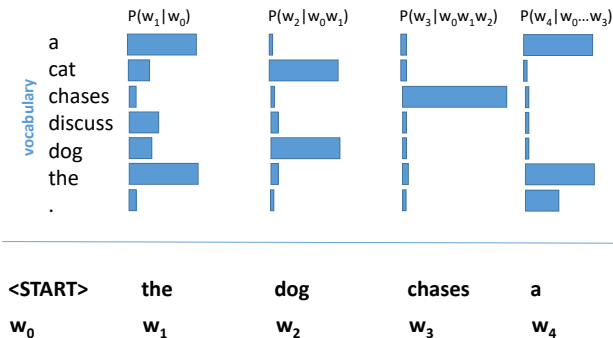
ARLM: TOY EXAMPLE



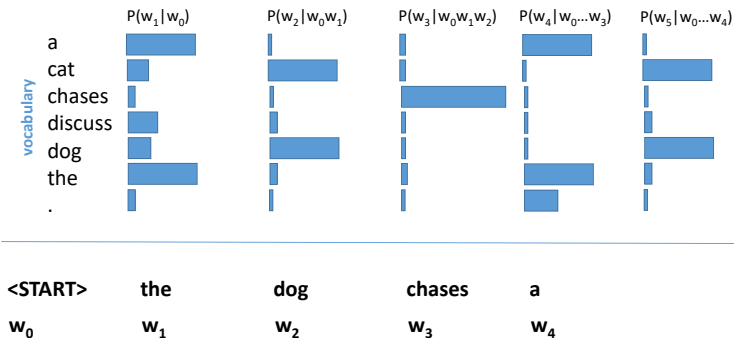
ARLM: TOY EXAMPLE



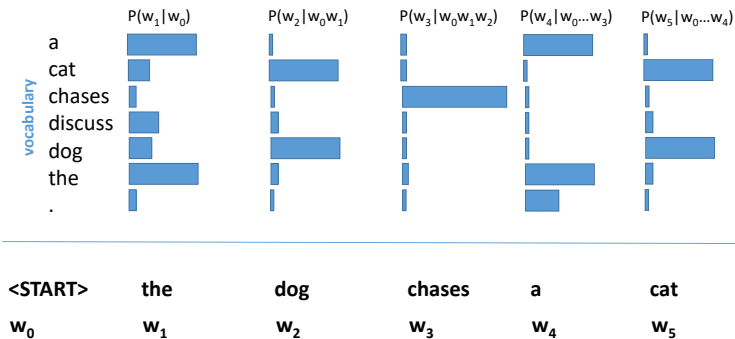
ARLM: TOY EXAMPLE



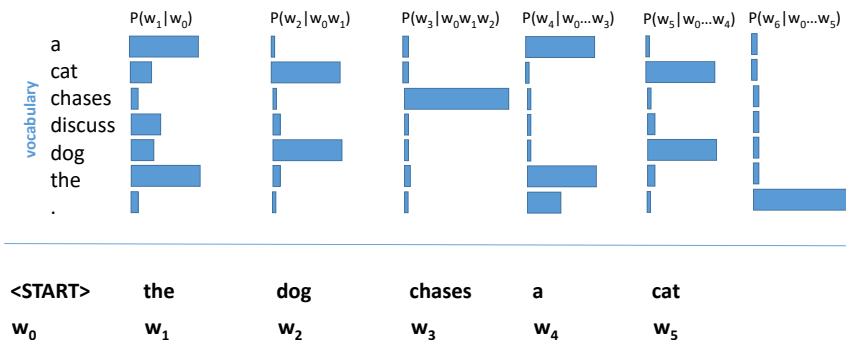
ARLM: TOY EXAMPLE



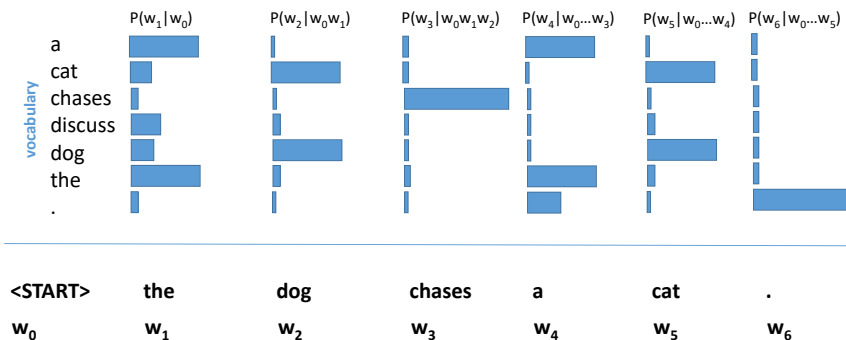
ARLM: TOY EXAMPLE



ARLM: TOY EXAMPLE



ARLM: TOY EXAMPLE



ARLM: PROBABILISTIC INTERPRETATION I

- Gives an estimate for the probability of a sentence using conditional probabilities
- In general:

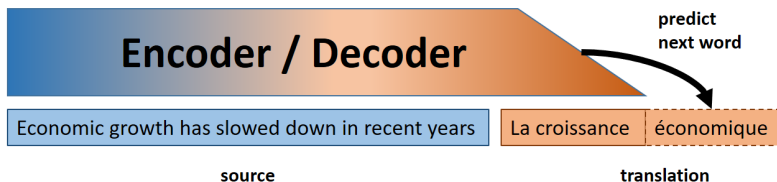
$$P(A \cap B) = P(B) \cdot P(A|B)$$

- For $P(\text{sentence})$:

$$\begin{aligned} &P(w_1, w_2, \dots, w_n | w_0) \\ &= P(w_1 | w_0) \cdot P(w_2 | w_0, w_1) \cdot \dots \cdot P(w_n | w_0, \dots, w_{n-1}) \end{aligned}$$

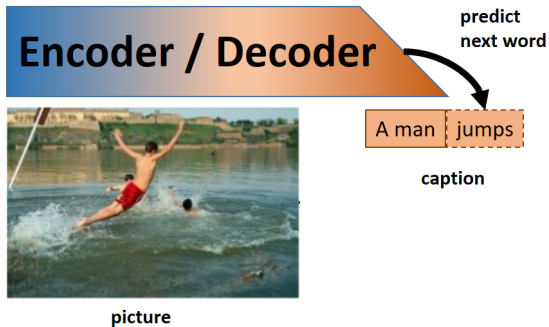
EXAMPLES OF ARLMS (1) I

Neural Machine Translation:



EXAMPLES OF ARLMS (1) I

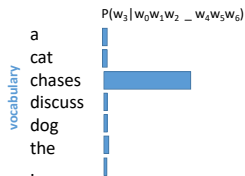
Image Captioning:



MASKED LANGUAGE MODELS (MLM) I

- We have seen auto-regressive LMs
 - context: previous words
 - predict: next word
- Another type: *Masked* LMs (MLMs)
 - context: surrounding words
 - predict: masked word

MLM: TOY EXAMPLE I



<START>	the	dog	<MASK>	a	cat	.
w_0	w_1	w_2		w_4	w_5	w_6

12

MLM: PROBABILISTIC INTERPRETATION I

- Estimates $P(w_i | w_0, w_1, w_{i-1}, \dots, w_{i+1}, w_n)$
- No “clean” estimate for $P(sentence)$, as

$$P(w_1, w_2, \dots, w_n | w_0)$$

$$\neq P(w_1 | w_0, w_2, \dots, w_n) \cdot P(w_2 | w_0, w_1, w_3, \dots, w_n)$$

- ARLMs are better than MLMs for generating texts
- Advantage of MLMs: Learning contextualized representations

Self-supervised Learning



DEFINITION

Unsupervised Learning:

- No labels attached to the data
- Learn patterns / clusters from the features only

Supervised Learning:

- (Gold) Labels attached to the data
- Learn from the association between features and labels

Self-Supervised Learning:

- No *external* labels attached to the data
→ Samples with suitable labels can be generated from the known structure of the data itself
- *Technically* supervised learning, but *no labeling effort* + simultaneous ability to generate massive amounts of labeled data points

SELF-SUPERVISED OBJECTIVES I

Self-supervised objectives:

- Skip-gram objective (cf. word2vec)
- Language modeling objective
- *Masked language modeling (MLM)* objective
- ... and many more possibilities for text data